1 Abstract

Classical *Schrödinger Bridge* (SB) problem seeks a path measure that minimally deviates from a given reference measure subject to prescribed initial and terminal marginals. This works investigate numerical approach to dynamic of interacting brownian particle with normal reflection term in low-dimensional setting ($d \leqslant 3$) on bounded domain $\mathcal{D} \subset \mathbb{R}^d$. To alleviate the difficulty due to nonlinearity incurred by non-local coupling, a nested fixed-point is adopted with Sinkhorn-type iterates. Further, fully semi-lagrangian discretization is used to propagate the solution of forward-backward partial differential equations (FB-PDEs).

2 Introduction

The random motion of microscopic particles, such as gas molecules or pollen grains suspended in a fluid, can be modeled by Brownian motion. When the number of particles is asymptotically large, it is natural to analyze their collective behavior via the evolution of their empirical distribution rather than tracking each individual path. Beyond merely predicting the passive evolution of a particle cloud, one often seeks to actively control it toward a desired target distribution—with applications ranging from particle beam focusing and swarm robotics to molecular transport and nonequilibrium thermodynamics.

A Brief History of the Schrödinger Bridge Problem Suppose the particle distribution is dictated by free Brownian diffusion, but an observed terminal distribution differs significantly from the theoretical diffusion prediction. Such an observation constitutes a rare event, raising a fundamental question: *which stochastic evolution is the most probable one?* This question lies at the heart of the *Schrödinger Bridge* (SB) problem. In his seminal 1931 work *Über die Umkehrung der Naturgesetze* [**?**], Erwin Schrödinger posed:

> *"Imagine that you observe a system of diffusing particles in thermal equilibrium. Suppose at time t_0 their distribution is more or less uniform and that at a later time $t_1 > t_0$ you find a significant spontaneous deviation from this uniformity. You are asked to explain how this deviation occurred. What is its most likely behavior?"*

Using the large deviation principle, Föllmer [**?**] rigorously formulated this heuristic in a modern measure-theoretic framework as a relative entropy minimization between path measures subject to prescribed initial and endpoint marginals, recovering the classical Schrödinger Bridge minimization:

$$\mathcal{J}^*_{\mathrm{SB}} :- \inf_{\substack{\mathbb{Q}\in\mathscr{P}(\Omega) \\ \mathbb{Q}_0=\mu_{\mathrm{init}},\, \mathbb{Q}_T=\mu_{\mathrm{targ}}}} H[\mathbb{Q} \parallel \mathbb{P}] \tag{1}$$

A related static interpolation problem appeared centuries earlier in Gaspard Monge's 1781 formulation of optimal transport (OT). By the Benamou–Brenier dynamic formulation [**?**], SB can be understood as the entropic regularization of dynamic OT [**?** **?**]. In the vanishing diffusion limit ($\kappa \to 0$), the solution of the SB problem converges directly to displacement interpolation in dynamic OT [**?** **?**].
Furthermore, (1) can be recast as a stochastic optimal control (SOC) problem subject to controlled stochastic differential equations (SDEs) [**?** **?** **?**]. The associated optimality system links a backward Hamilton–Jacobi–Bellman (HJB) PDE for the value function with a forward Fokker–Planck (FP) PDE for the probability density, yielding a coupled system of forward-backward PDEs (FB-PDEs).

From Classical SB to Reflected Mean-Field Dynamics Crucially, classical SB presumes that particles evolve freely across the entire Euclidean space $\mathbb{R}^d$ without spatial or geometric constraints. However, in many physical scenarios—from molecular transport within cellular organelles to colloidal suspensions in microfluidic channels—the admissible state space is bounded and restricted by physical boundaries. Standard Brownian motion is no longer applicable, as unconstrained trajectories would exit the physical domain. To resolve this, one must incorporate normal boundary reflections, modeling the underlying dynamics as a reflected Itô-diffusion governed by McKean–Skorokhod SDEs on a bounded domain $\mathcal{D} \subset \mathbb{R}^d$.
At the same time, in interacting particle systems and thermodynamic ensembles, microscopic particles often interact through non-local spatial forces (e.g., inter-particle attraction, steric repulsion, or pair potentials). This gives rise to the Mean-Field Schrödinger Bridge (MFSB) problem [**?** **?**], where the state dynamics depend on non-local spatial convolutions ($\nabla E \star \rho_t$). Combining spatial reflection constraints with non-local mean-field interactions yields the Reflected Mean-Field Schrödinger Bridge (MFRSB) problem—a highly non-linear, non-convex system of forward-backward PDEs with no-flux Neumann boundary conditions that demands specialized numerical treatment.

Beyond its physical foundations, the Schrödinger Bridge framework has recently gained significant traction in machine learning, generative modeling, biological cell trajectory inference, and multi-agent swarm control [**? ? ?**].

3 Related Works

Interest in spatially-constrained SB is largely under-developed, especially when also incorporating mean-field framework. This work is inspired by [4, 11]. First paper studies SB with reflected dynamic with Sinkhorn iterate where the time marching solution involve Neuman and Rudin boundary conditions, and show great performance in low-dimension. The latter generalized standard Sinkhorn to accept partial-integro-PDEs associated with mean-field SB resulting in nested version. Along this line [1] studies degenerate, sub-Riemannian dynamics, and use Sinkhorn simiarl to the former. Recent work [16] use Finite element discretization to solve the weak form of coupled PDEs for reflected SB. [9] derives forward-backward SDEs associated coupled FB-PDEs of reflected SB and extend divergence-based likelihood training in smooth bounded domain and shown to scale in high-dimension. Discrete-time framework also been explored with chance-constrants [18, 19] with mixture of Guassian marginal contraints. Alternative method such as introducing direct soft-penalty for particle escaping region exists as demonstrated in [15] using FB-SDE representation. There is a direct link between system of optimality of SB and MFG, therefore one can naturally adopt a scheme in MFG such as finite-difference to solve SB problem. [5–7] adopt a semi-lagrangian approach.

4 Problem Formulation

Let $\mathcal{D} \subset \mathbb{R}^d$ and boundary $\partial\mathcal{D}$, represented by smooth level-set $\chi(t,x)$ via $\chi < 0$ on $\mathcal{D}$ and $\chi = 0$ on $\partial\mathcal{D}$. Assuming $\nabla\chi \neq 0$ on $\partial\mathcal{D}$, the outward unit normal $\mathbf{n}(x)$ is obtained by $\mathbf{n} = \nabla\chi/\|\nabla\chi\|$. Denote by $\mathscr{P}(\mathcal{S})$ the space of probability measures on $\mathcal{S}$ endowed with the 2-Wasserstein metric $W_2(\cdot,\cdot)$ and having finite 2nd-moment. Let $\Omega :- \mathcal{C}(\mathcal{T};\mathbb{R}^d)$ where $\mathcal{T} :- [0,T]$ be canonical path space. Let $\mathbb{P}_t \in \mathscr{P}(\mathbb{R}^d)$ be t-time pushforward marginal of $\mathbb{P} \in \mathscr{P}(\Omega)$. For given fixed $\mathbb{P}$ and $\mu_{\text{init}} \in \mathscr{P}(\mathbb{R}^d)$, consider the dynamic of weakly-interacting brownian particle modeled by Mckean-Vlaskov reflected SDEs:

$$dX_t = -(\nabla E \star \mathbb{P}_t)(X_t) + \sigma\ dW_t - \mathbf{n}_t dV_t \qquad , X_0 \sim \mu_{\text{init}} \tag{2}$$

which induces the path law $\Gamma(\mathbb{P}) :- \text{Law}\left((X_t)_{t\in\mathcal{T}}\right)$, W is $\mathbb{R}^d$-valued Wiener on $(\Omega, \mathcal{F}, \mathbb{F} = (\mathcal{F}_t)_{t\in\mathcal{T}}, \mathbb{P}_0)$ Here E is potential, $(\nabla E \star \mu)(\cdot) :- \int_{\mathbb{R}^d} \nabla E(\cdot - y)\ \mu(dy)$. And V is a continuous, non-decreasing, $\mathbb{F}$-adapted bounded-variation process satisfying $V_0 = 0$, $\int_0^T \mathbf{1}_{\{X_s \notin \partial\mathcal{D}_s\}}\ dV_s = 0$, and $V_t = \int_0^t \mathbf{1}_{\{X_s \in \partial\mathcal{D}_s\}}\ dV_s$. For simplicity takes diffusion be $\sigma \equiv \sqrt{2\kappa}\mathbb{I}_d$ for $\kappa_0 > 0$. In [] shows that the rate function governing the large deviations for mean-field type dynamic in (1) is $H(\mathbb{P} \parallel \Gamma(\mathbb{P}))$. Unlike standard non-mean-field SB in that the referenc measure is no longer fixed and the convexity property is losed due to nonlinear dependency in its arguments of relative entropy []. Under appropriate hypotheses, the existence of $\mathbb{F}$-progressively measurable $\alpha(t,x)$ is guaranteed by Girsanov's, the relative entropy can be parametrized by process α:

$$H\big(\mathbb{P} \parallel \Gamma(\mathbb{P})\big) = \frac{1}{2}\mathbb{E}^{\mathbb{P}}\left[\int_0^T \|\alpha(t,X_s)\|^2\ ds\right] \tag{3}$$

and that X_t follows the controlled Mckean Vlasov SDE:

$$dX_t = b^\alpha(t, X_t, \mathbb{P}_t)\ dt + \sigma_t\ dW_t^{\mathbb{P}} - \mathbf{n}_t dV_t, \quad X_0 \sim \mu_{\text{init}}, \tag{4}$$

where $W^{\mathbb{P}}$ is a standard $(\mathbb{P}, \mathcal{F}_t)$-Brownian, $b^\alpha(t,x,\mu) :- \sigma_t\alpha_t - \nabla E \star \mu$. Further, under additional hypothesis, the perturbation is markovian feedback control. i.e. there exists $\Phi_t \in H_{-1}(\rho_t : t \in \mathcal{T})$ such that $\alpha_t = \Phi_t(X_t)$, $dt \times d\mathbb{P}$ a.s. (here assuming the control dimension is similar to state dimension). Following [], using transformed control $m := \rho\alpha \in \mathbb{R}^d$, rewrite (2) as SOC problem

$$\mathcal{J}^*_{\text{rmf-sb}} = \inf_{(\alpha,\rho)} \mathcal{J}_{\text{rmf-sb}}(m,\rho), \text{ where } \mathcal{J}_{\text{rmf-sb}}(m,\rho) :- \frac{1}{2}\int_0^T \int_{\mathbb{R}^d} \|m_t\|^2/\rho_t dx\ dt. \tag{5}$$

The law of sample path $(X_t)_{t\in\mathcal{T}}$ in (3) can be characterized as solving the martingale problem, i.e. under suitable assumption, for any $\psi \in \mathcal{C}^2(\mathcal{D})$, the process

$$M_t :- \psi(X_t) - \psi(X_0) - \int_0^t \mathcal{G}[\rho_s]\psi(X_s)ds \tag{6}$$

is $\mathbb{P}$-local martingale with respect to $\mathbb{F}$ where $\mathcal{G}$ is generator for (3) defined by $\mathcal{G}[\rho] := \kappa\nabla\rho + \nabla\rho \cdot (\sigma\alpha - E \star \rho)$. It follows that [], the marginal flows $(\rho_t)_{t\in\mathcal{T}}$ is a weak solution of forward Kolmogorov equation:

$$\begin{aligned} \partial_t\rho + \nabla \cdot J = 0, \ (t,x) \in \mathcal{D}, \\ J \cdot \mathbf{n} = 0, \ (t,x) \in \partial\mathcal{D} \end{aligned} \tag{7}$$

where probability flux is $J(m,\alpha) = \sqrt{2\kappa}m - \rho\nabla E \star \rho - \kappa\nabla\rho$, where here using $m := \rho\alpha$. One can now characterize the 1st-order optimality condition associated with (4) with contraint (6) which yields a system of forward-backward PDEs. Here the system is adjacent to that appear in [], supplemented with additional Neuman and Robin boundary conditions. The proof given here is straightforward application of Lagrange mulitplier on Banach space.

4.1 Assumption

- The domain $\mathcal{D} \subset \mathbb{R}^d$ is sufficiently regular.
- Interacting term $E \in \mathcal{C}^2\left(\mathbb{R}^d;\mathbb{R}\right)$ is symmetric, i.e. $E(\cdot) = E(-\cdot)$ with $\sup_{z,v\in\mathbb{R}^d} v^\top\nabla^2 E(z)v < +\infty$.
- The admissible controls $\alpha \in \Theta_\alpha := H^1(\rho)$ s.t. $\exists\psi \in H^1$ with $\alpha = \nabla\psi$, $|m|^2/\rho \in L^1$, and $\rho \in \mathcal{C}\left(\mathcal{T}, \mathscr{P}(\mathcal{D})\right) \cap L^2\left(0,T;H^1(\mathcal{D})\right)$ with $\rho \geqslant 0$, $\int_{\mathcal{D}}\rho = 1$ and $\rho(0,\cdot)dx = \mu_{\text{init}}$, $\rho(T,\cdot)dx = \mu_{\text{targ}}$

The rest of this work focus solving the following coupled FB-PDEs which can be drived in straightforward manner by mean of Lagrange Multiplier on Banach space. The difference between (PDE-opt) and that which given by [2] is the extra boundary condition.

Proposition 1. *[2] The optimal control α^*, ρ^* are given by $\alpha^* = \sqrt{2\kappa}\nabla\log\varphi$ where $(\phi,\rho) \to (\varphi,\hat{\varphi}) := (e^{-\phi}, \rho\exp(E\star\rho/\kappa + \phi)$ solve the following FB-PDEs:*

$$\begin{cases} -\partial_t\varphi = \left(\mathcal{A}^\rho + \mathcal{R}^\rho\right)[\varphi] \\ \partial_t\hat{\varphi} = \left(\mathcal{A}^{\rho,\dagger} + \mathcal{R}^{\rho,\dagger}\right)[\hat{\varphi}] \end{cases}, \text{ on } (0,T)\times\mathcal{D}, \quad \text{such that} \begin{cases} \nabla\varphi^\top\mathbf{n} = \nabla\hat{\varphi}^\top\mathbf{n} = 0, \ \text{on } (0,T)\times\partial\mathcal{D} \\ \mu_{init} = W[\rho]\varphi_0\hat{\varphi}_0, \ \text{on } \{0\}\times\mathcal{D} \\ \mu_{targ} = W[\rho]\varphi_T\hat{\varphi}_T, \ \text{on } \{T\}\times\mathcal{D} \end{cases} \tag{8}$$

where $\mathcal{A}^\rho[\eta] := (\nabla E\star\rho)\cdot\nabla\eta - \kappa\Delta\eta$ is advection-diffusion operator and $\mathcal{R}^\rho[\eta] := -\eta\nabla E\star(\rho\nabla\log\eta)$ is nonlocal source term, and $W[\rho](t,x) := e^{-E\star\rho/\kappa}$.

4.2 Fixed-Point Iterates

The constructive scheme for the PDE-opt solver is obtained via a single-loop Picard-Sinkhorn iteration to find the fixed point of the flow $\rho = (\rho_t)_{t\in\mathcal{T}} \in \mathcal{C}(\mathcal{T}\mathscr{P}(\mathcal{D}))$. We define the marginal map F and the inner propagator G by

$$F[\rho;\varphi,\hat{\varphi}] := W[\rho]\varphi\hat{\varphi} \quad \text{and} \quad G[\varphi_T;\mu,\varphi,\hat{\varphi}] := \left(M_T^\mu \circ \mathcal{S}_{0\to T}^{\dagger,\mu,\hat{\varphi}} \circ M_0^\mu \circ \mathcal{S}_{T\to 0}^{\mu,\varphi}\right)[\varphi_T]. \tag{9}$$

The propagator G shares a very similar structure with the classical linear Schrödinger bridge (SB), which is a well-studied numerical method [8, 14]. Here, $\mathcal{S}_{0\to T}^{\dagger,\mu,\hat{\varphi}}$ and $\mathcal{S}_{T\to 0}^{\mu,\varphi}$ are respectively the backward and forward time-split propagators of the linearized PDE-opt system frozen at $(\mu,\varphi,\hat{\varphi})$. The term "linearized PDEs" refers to the fact that under fixed $(\rho^{n-1},\varphi^{n-1},\hat{\varphi}^{n-1})$, both the nonlocal drift $b_t^{n-1} := -\nabla E * \rho_t^{n-1}$ and the reaction rate $r_t^{n-1} := -\nabla E * (\rho_t^{n-1}\nabla\log\varphi_t^{n-1}) \in L^\infty(\mathcal{D})$ are frozen in time and space, reducing the nonlinear forward-backward PDEs to standard non-autonomous linear Cauchy problems with the time-varying generator $\mathcal{A}_t^{n-1} + \mathcal{R}_t^{n-1}$. It follows that this generator induces a unique backward evolution family $(\mathcal{P}_{t\to s}^{n-1})_{0\leqslant s\leqslant t\leqslant T}$ on $L^2(\mathcal{D})$; see also [3]. Here, M_T^μ and M_0^μ are Sinkhorn scaling operators defined pointwise by

$$\hat{\varphi}_0(\cdot) \xleftarrow{M_0^\mu} \frac{\mu_{\text{init}}(\cdot)}{W[\mu](\cdot,0)\varphi(\cdot,0)} \quad \text{and} \quad \hat{\varphi}_T(\cdot) \xleftarrow{M_T^\mu} \frac{\mu_{\text{targ}}(\cdot)}{W[\mu](\cdot,T)\varphi(\cdot,T)}. \tag{10}$$

Note that these two operators are similar to those in [11]. We suppress the superscripts ρ^{n-1},φ^{n-1} for readability. For a uniform temporal subdivision $0 = t_0 < t_1 < \cdots < t_{N_T} = T$ of step size $\Delta t = t_{k+1} - t_k$, we define the exact continuous symmetric Strang splitting $\mathcal{S}_{t_{k+1}\to t_k}$ acting on φ via the composition

$$\varphi(t_k,\cdot) \approx \mathcal{S}_{t_{k+1},t_k}[\varphi](t_{k+1},\cdot) := \left(\mathcal{R}_{t_{k+\Delta t/2}\to t_k} \circ \mathcal{U}_{t_{k+1}\to t_k} \circ \mathcal{R}_{t_{k+1}\to t_{k+\Delta t/2}}\right)[\varphi](t_{k+1},\cdot). \tag{11}$$

Since $\mathcal{R}[\eta](s,x) = r_s(x)\eta(x)$ is a bounded pointwise multiplication operator on $L^2(\mathcal{D})$, it admits an analytical solution of the form $\mathcal{R}_{t\to s}[\eta](x) = \eta(x)\exp\int_s^t r_u(x)\,du$, whose half-step approximations via quadrature points are given by

$$\tilde{\mathcal{R}}_{k+1\to k+\Delta t/2}[\eta] := \eta e^{\Delta t r_{t_{k+1}}/2} \quad \text{and} \quad \tilde{\mathcal{R}}_{k+\Delta t/2\to k}[\eta] := \eta e^{\Delta t r_{t_k}/2}. \tag{12}$$

Next, the middle operator $\mathcal{U}_{t_{k+1}\to t_k}$ solving the advection-diffusion subproblem has the Feynman-Kac representation [17] over $[t_k, t_{k+1}]$ with Neumann boundary conditions given by

$$\mathcal{U}_{t_{k+1},t_k}[\eta](x) := \mathbb{E}^{\mathbb{P}}\left[\eta(t_k, X_{t_k}) \mid X_{t_{k+1}} = x\right] = \lim_{K\to\infty} \sum_{1\leqslant m\leqslant K} w_m \eta\left(t, X^x_{t_k,t_{k+1}}(\omega_m)\right), \tag{13}$$

where $X_{t_k,t_{k+1}}$ solves (for fixed ρ) the forward process starting from $X_{t_k} = x$, and w_m are normalized positive weights. Due to random noise incurred by finite approximations of expectations via Monte Carlo (of order $O(\sqrt{\Delta t/K})$), a deterministic cubature stencil is used instead, known as the Semi-Lagrangian (SL) method. Defining $\mathcal{S}_{T\to 0} := \mathcal{S}_{t_1\to t_0} \circ \mathcal{S}_{t_2\to t_1} \circ \cdots \circ \mathcal{S}_{t_{N_T}\to t_{N_{T-1}}}$, it is guaranteed [3] that a second-order error bound for sufficiently smooth terminal data φ_T holds; i.e., there exists $C \geqslant 0$ independent of Δt such that

$$\|\mathcal{P}_{0,T}\varphi_T - \mathcal{S}_{T\to 0}\varphi_T\|_{L^2} \leqslant C\Delta t^2 \|\varphi_T\|_{L^2}. \tag{14}$$

Similarly, since the Fokker-Planck equation for $\hat{\varphi}$ is the formal L^2-adjoint of the backward equation, $\mathcal{S}^\dagger_{k\to k+1} := (\mathcal{S}_{k+1\to k})^\dagger$ is obtained simply by reversing the operation:

$$\hat{\varphi}(t_{k+1},\cdot) \approx \mathcal{S}^\dagger_{t_k\to t_{k+1}}[\hat{\varphi}](t_k,\cdot) = \left(\mathcal{R}^\dagger_{t_{k+1}\to t_{k+\Delta t/2}} \circ \mathcal{U}^\dagger_{t_{k+1}\to t_k} \circ \mathcal{R}^\dagger_{t_{k+\Delta t/2}\to t_k}\right)[\hat{\varphi}](t_k,\cdot). \tag{15}$$

Notice that $\mathcal{R}^\dagger_{t\to s} = \mathcal{R}_{t\to s}$ is self-adjoint. Upon spatial discretization, this structure ensures the discretized forward propagator $\mathcal{S}_{0\to T}$ can be obtained via mass-weighted matrix transpose of $\mathcal{S}_{T\to 0}$; see also [12].

Algorithm 1: Fixed-Point Iteration Solver

Input: $\mu_{\text{init}}, \mu_{\text{targ}} \in \mathbb{R}^{N_x}_{>0}$, $\Delta t = T/N_T$, $\theta \in (0,1]$, threshold $\epsilon > 0$.
Initialize: $\boldsymbol{\varphi}^0_k, \hat{\boldsymbol{\varphi}}^0_k \leftarrow \mathbb{1}_{N_x}$ and $\boldsymbol{\rho}^0 \leftarrow \mathbb{1}_{N_x}/\operatorname{Vol}_d(\mathcal{D})$ for $k = 0,\ldots,N_T$. **for** $n = 1,\ldots,N_{iter}$ **do**
  Compute $\mathbf{V}^{n-1}_k, \mathbf{W}^{n-1}_k, \mathbf{R}^{n-1}_k, \mathbf{r}^{n-1}_k, \mathbf{b}^{n-1}_{k,l}$ for $k = 0,\ldots,N_T$
  Set $\boldsymbol{\varphi}^n_{N_T} \leftarrow \boldsymbol{\varphi}^{n-1}_{N_T}$
  for $k = N_T-1, N_T-2, \ldots, 0$ **do**
    $\boldsymbol{\varphi}^n_k \leftarrow \mathbf{R}^{n-1}_k \boldsymbol{\Pi}^{n-1}_k \mathbf{R}^{n-1}_{k+1} \boldsymbol{\varphi}^n_{k+1}$
  Set $\hat{\boldsymbol{\varphi}}^n_0 \leftarrow \boldsymbol{\mu}_{\text{init}} \oslash \left(\operatorname{diag}(\mathbf{W}^{n-1}_0) \odot \boldsymbol{\varphi}^n_0\right)$
  for $k = 0,\ldots,N_T-1$ **do**
    $\hat{\boldsymbol{\varphi}}^n_{k+1} \leftarrow \mathbf{R}^{n-1}_{k+1} \boldsymbol{\Pi}^{n-1,\dagger}_k \mathbf{R}^{n-1}_k \hat{\boldsymbol{\varphi}}^n_k$
  Set $\boldsymbol{\varphi}^n_{N_T} \leftarrow \boldsymbol{\mu}_{\text{targ}} \oslash \left(\operatorname{diag}(\mathbf{W}^{n-1}_{N_T}) \odot \hat{\boldsymbol{\varphi}}^n_{N_T}\right)$
  Compute $\boldsymbol{\rho}^{1/2}_k = \operatorname{diag}(\mathbf{W}^{n-1}_k) \odot \boldsymbol{\varphi}^n_k \odot \hat{\boldsymbol{\varphi}}^n_k$ and normalize in $L^1(\mathcal{D}_\Delta)$: $\boldsymbol{\rho}^{1/2}_k \leftarrow \boldsymbol{\rho}^{1/2}_k / (\mathbf{1}^\top M_{\text{lump}} \boldsymbol{\rho}^{1/2}_k)$
  Set $\boldsymbol{\rho}^n_k = (1-\theta)\boldsymbol{\rho}^{n-1}_k + \theta \boldsymbol{\rho}^{1/2}_k$ for $k = 0,\ldots,N_T$
  if $\max_{1\leqslant k\leqslant N_T-1} \|\boldsymbol{\rho}^n_k - \boldsymbol{\rho}^{n-1}_k\|_\infty \leqslant \epsilon$ **then**
    break

Unlike the nested two-loop scheme in [10, 12, 13], Solver 1 executes only a single loop, and the nonlocal drift and reaction terms are updated alongside the scaling operators M_0 and M_T. This structure closely mirrors fictitious play and Picard iteration in mean-field games (MFG) [**?**] with extended Sinkhorn scaling applied to PDEs. The next section concerns spatial discretization via a monotone P_1 finite element Semi-Lagrangian scheme.

Algorithm 2: Solver 1: Fixed-Point Iteration

for $n = 1,\ldots,N_{iter}$ **do**
  $\varphi^n \leftarrow \mathcal{S}_{T\to 0}\left[\varphi^{n-1}_T; \rho^{n-1}, \varphi^{n-1}\right]$ `// backward propagate`
  $\hat{\varphi}^n_0 \leftarrow M_0^{\rho^{n-1}}[\varphi^n]$ `// initial marginal scale`
  $\hat{\varphi}^n \leftarrow \mathcal{S}^\dagger_{0\to T}[\hat{\varphi}^n_0; \rho^n, \hat{\varphi}^n]$ `// forward propagate`
  $\varphi^n_T \leftarrow M_T^{\rho^{n-1}}[\hat{\varphi}^n]$ `// terminal marginal scale`
  $\rho^n \leftarrow (1-\theta)\rho^{n-1} + \theta F\left[\rho^{n-1}; \varphi^n, \hat{\varphi}^n\right]$ `// density update`

4.3 Semi-Lagrangian Discretization

What remains now is spatial discretization via SL. As mentioned in [5], the SL scheme is monotone, unconditionally stable, and positivity-preserving without requiring the parabolic CFL condition $\Delta t = O(\Delta x^2)$. Let

$\mathcal{D}_\Delta = \{x_i\}_{1\leqslant i\leqslant N_x} \subset \overline{\mathcal{D}}$ be the vertices of a simplicial triangulation of size Δx over the polyhedral domain $\mathcal{D}$, and let $\mathcal{T}_\Delta = \{t_k = k\Delta t\}_{0\leqslant k\leqslant N_t}$ be the time subdivision. At t_k, the continuous field $\eta \in L^2(\mathcal{D})$ is approximated by nodal values $\boldsymbol{\eta}_k = (\eta_{k,1}, \ldots, \eta_{k,N_x}) \in \mathbb{R}^{N_x}$. Its continuous reconstruction is obtained via the P_1 finite element interpolator $\mathcal{I} : \mathbb{R}^{N_x} \to \mathcal{C}_b(\mathcal{D})$ defined by $\mathcal{I}[\eta](\cdot) :- \boldsymbol{\beta}(\cdot)^\top \boldsymbol{\eta}$, where $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_{N_x})^\top$ is the vector of standard piecewise linear Lagrange basis functions satisfying $\beta_i(x_j) = \delta_{ij}$ and $\mathbf{1}^\top \boldsymbol{\beta} = 1$. This interpolation introduces a spatial discretization error strictly bounded by $O(\Delta x^2)$. Following [7], we choose $\beta_i(x) = \max\{0, 1 - |x - x_i|/\Delta x\}$, which has support on $[x_i - \Delta x, x_i + \Delta x]$. The choice of linear interpolants (yielding $O(\Delta x^2)$ error) over higher-order alternatives such as Hermite polynomials or splines (which achieve spatial errors up to $O(\Delta x^4)$) is motivated by properties endowed by $\mathcal{I}$, such as positivity preservation, monotonicity, and computational efficiency.

We approximate $\mathcal{U}_{t_{k+1}\to t_k}$ with a second-order weak $(2d+1)$-point stencil; see [5]. For node $x_i \in \mathcal{D}_\Delta$ at time t_k, the advective transport under reference drift $\mathbf{b}_{k,i}^{n-1} :- (-\nabla E \star \rho_k^{n-1})(x_i) \in \mathbb{R}^d$ with diffusivity $2\kappa\mathbb{I}_d$ is approximated by the unconstrained backward characteristic path over $[t_k, t_{k+1}]$ via

$$y^s_{k,i,l}(a) = x_i - \Delta t \mathbf{b}_{k,i}^{n-1} + e_l s\sqrt{6\kappa d\Delta t}, \quad s \in \{0, \pm 1\},\ l \in \{1, \ldots, d\}, \tag{16}$$

where e_l is the canonical basis vector. Neumann reflection boundary conditions are enforced for any characteristic exiting $\mathcal{D}$, following [5] for the special case of normal reflection. Specifically, recalling that $\chi < 0 \iff x \notin \mathcal{D}$ denotes the signed distance function, the Skorokhod map reads

$$\hat{y}^s_{k,\mathbf{i},\ell} :- \begin{cases} y^s_{k,\mathbf{i},\ell} & \text{if } y^s_{k,\mathbf{i},\ell} \in \mathcal{D}, \\ y^s_{k,\mathbf{i},\ell} - 2\chi(y^s_{k,i,l})\mathbf{n}(\chi(y^s_{k,i,l})) & \text{if } y^s_{k,\mathbf{i},\ell} \notin \mathcal{D}, \end{cases} \tag{17}$$

where $\mathbf{n}(\chi(a)) = \nabla\chi(a)/\|\nabla\chi(a\|_2$. Using these positions and three stencil weights, the fully discrete SL operator approximated at x_i is given by the transition matrix $\boldsymbol{\Pi}_k \in \mathbb{R}^{N_x\times N_x}$ via $\mathcal{U}_{t_{k+1}\to t_k}[\boldsymbol{\eta}](x_i) \approx \sum_{1\leqslant j\leqslant N_x} (\boldsymbol{\Pi}_k^{n-1})_{i,j}\eta_j$, where

$$(\boldsymbol{\Pi}_k)_{a,b} :- \frac{2}{3}\beta_j(\hat{y}^0_{k,i,1}) + \frac{1}{6d}\sum_{\ell=1}^{d}\left(\beta_j(\hat{y}^+_{k,i,\ell}) + \beta_j(\hat{y}^-_{k,i,\ell})\right). \tag{18}$$

This choice achieves second-order accuracy. For the increment $\Delta Y_{k,i} = Y_{k,i} - x_i$, we have $\mathbb{E}[\Delta Y_{k,i}] = -\Delta t \mathbf{b}_k^{n-1}(x_i)$ and $\mathrm{Cov}(\Delta Y_{k,i}) = 2\kappa\Delta t\mathbb{I}_d + O(\Delta t^2)$, which is congruent with $\mathbb{E}[dX_t]$ and $\mathrm{Cov}(\sqrt{2\kappa}dW_t)$ up to second order, respectively. Furthermore, by construction, the matrix $\boldsymbol{\Pi}$ is row-stochastic (i.e., non-negative with $\boldsymbol{\Pi}^\top\mathbf{1} = \mathbf{1}$). For a probabilistic interpretation of the SL scheme in terms of controlled Markov processes, see [5]. Finally, let $M_{\mathrm{lump}} = \mathrm{diag}(m_1, \ldots, m_{N_x})$ with mass weights $m_i = \int_{\mathcal{D}} \beta_i(a)\, da > 0$, and let $\mathbf{D}_l \in \mathbb{R}^{N_x\times N_x}$ be the lumped-mass discretized partial derivative ∂_{x_l} along $l \in \{1, \ldots, d\}$ defined by $(\mathbf{D}_l)_{i,j} = m_i^{-1}\int_{\mathcal{D}} \beta_i(x)\partial_{x_l}\beta_j(x)\, dx$.

The discretized objects are summarized below for iteration $n-1$ (dropping superscripts for clarity), step k, $l \in \{1, \ldots, d\}$, and $i \in \{1, \ldots, N_x\}$:

$$\begin{aligned} (\mathbf{V}_k)_i &:- \sum_j E(x_i - x_j)\rho_{k,j}m_j, & \mathbf{W}_k &:- \mathrm{diag}\left(\exp(-\mathbf{V}_k/\kappa)\right), \\ \mathbf{R}_k &:- \mathrm{diag}\left(e^{\Delta t\mathbf{r}_k/2}\right), & \mathbf{r}_k &:- -\sum_{l,j}\rho_{k,j}m_j\partial_{x_l}E(x_i - x_j)^\top\frac{(\mathbf{D}_l\boldsymbol{\varphi}_k)_j}{\varphi_{k,j}}, \\ (\mathbf{b}_{k,l})_i &:- -\sum_j \partial_{x_l}E(x_i - x_j)\rho_{k,j}m_j. \end{aligned} \tag{19}$$

With respect to the lumped $L^2(\mathcal{D}_\Delta)$ inner product $\langle\mathbf{u}, \mathbf{v}\rangle_{M_{\mathrm{lump}}} :- \mathbf{u}^\top M_{\mathrm{lump}}\mathbf{v}$, the discrete adjoint of the single-step backward matrix operator $\mathbf{S}_{k+1\to k} :- \mathbf{R}_k\boldsymbol{\Pi}_k\mathbf{R}_{k+1}$ is given explicitly by

$$(\mathbf{S}_{k+1\to k})^\dagger = M_{\mathrm{lump}}^{-1}(\mathbf{S}_{k+1\to k})^\top M_{\mathrm{lump}} = \mathbf{R}_{k+1}\boldsymbol{\Pi}_k^\dagger\mathbf{R}_k, \tag{20}$$

where $\boldsymbol{\Pi}_k^\dagger :- M_{\mathrm{lump}}^{-1}\boldsymbol{\Pi}_k^\top M_{\mathrm{lump}}$ due to commutation with M_{lump}.

5 Results

Table 1: Spatio-temporal convergence of the Strang-split semi-Lagrangian solver with simultaneous refinement $\Delta t \propto h_{\max}$. Errors are computed against an ultra-fine reference solution ($h_{\mathrm{ref}} = 0.01, \Delta t_{\mathrm{ref}} = T/160$).

Δt	$h_{\max}$	N_{nodes}	$\mathrm{err}_{L^\infty}(\varphi)$	EOC	$\mathrm{err}_{L^2}(\varphi)$	EOC	$\mathrm{err}_{L^2}(\nabla\varphi)$	EOC	CPU (s)
$T/10$	0.160	412	–	–	–	–	–	–	–
$T/20$	0.080	1520	–	–	–	–	–	–	–
$T/40$	0.040	5890	–	–	–	–	–	–	–
$T/80$	0.020	23100	–	–	–	–	–	–	–

Table 2: Temporal convergence and empirical order of convergence (EOC) for Lie vs. Strang operator splitting on a fixed mesh ($h_{\max} = 0.04$).

	Lie Splitting ($\mathcal{O}(\Delta t)$)			**Strang Splitting** ($\mathcal{O}(\Delta t^2)$)			
Δt	$\mathrm{err}_{L^2}(\varphi)$	EOC	$\mathrm{err}_{\mathrm{adj}}$	$\mathrm{err}_{L^2}(\varphi)$	EOC	$\mathrm{err}_{\mathrm{adj}}$	CPU (s)
$T/10$	–	–	–	–	–	–	–
$T/20$	–	–	–	–	–	–	–
$T/40$	–	–	–	–	–	–	–
$T/80$	–	–	–	–	–	–	–

Table 3: Convergence of the outer Picard iteration for the mean-field coupling ($\Delta t = T/40, h_{\max} = 0.04, \theta = 0.65$).

Iter n	$\|\rho^n - \rho^{n-1}\|_{L^1}$	$q_n = \frac{\|\Delta\rho^n\|}{\|\Delta\rho^{n-1}\|}$	$\|\rho_0^n - \mu_0\|_{L^1}$	$\|\rho_T^n - \mu_T\|_{L^1}$	$\mathcal{W}_2(\rho_T^n, \mu_T)$	$\max_k \lvert \log Z_k \rvert$
1	–	—	–	–	–	–
3	–	–	–	–	–	–
5	–	–	–	–	–	–
10	–	–	–	–	–	–
15	–	–	–	–	–	–
25	–	–	–	–	–	–

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